Phase 2: Stability Analysis of the Acrobot

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Abstract. Sections I–V restate the Phase 1 model: the given Acrobot dynamics, its four zero-torque equilibria, their linearizations and the transfer functions from the elbow torque τ to both joint angles. Sections VI–IX are Phase 2. We determine the stability of every equilibrium from the transfer functions and from the state matrix, and the two agree. We confirm every conclusion with a PyBullet simulation of the nonlinear model. At the upright equilibrium we show that no static or PD feedback from a single joint angle can stabilize it, because of real right-half-plane zeros. A dynamic observer-based controller can, but only for perturbations of about 1° under the 20 N m torque limit.

I. Nonlinear Model

A. System and Parameters. Two-link pendulum (Fig. 1) with a passive shoulder and a motor at the elbow [1], [2].

- Angles: q_1 = link 1 from the downward vertical; q_2 = link 2 relative to link 1; $\varphi = q_1 + q_2$.
- State: $x = [q_1,\ q_2,\ \dot{q}_1,\ \dot{q}_2]^T$.
- Input: elbow motor torque τ, $|\tau| \le 20\,\mathrm{N\,m}$, the only quantity we command ($\tau = 0$ in this phase; set by a controller in later phases).
- $m_1 = m_2 = 1$ kg, $l_1 = 1$ m, $l_2 = 2$ m, $l_{c1} = 0.5$ m, $l_{c2} = 1$ m, $I_1 = 0.083$, $I_2 = 0.33\ \mathrm{kg\,m^2}$, $g = 9.81\ \mathrm{m/s^2}$.

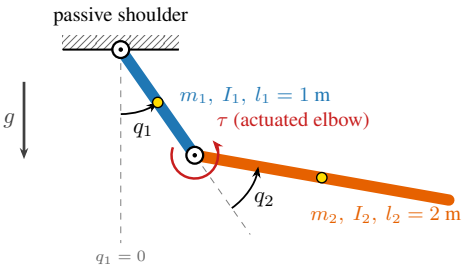


Fig. 1. Acrobot schematic (to scale); dots mark the centres of mass.

B. Given Equations of Motion.

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} = \tau_g(q) + B\tau, \qquad B = [0,\ 1]^T \tag{1}$$

(B applies τ to the elbow equation only; $c_2 = \cos q_2$, $s_2 = \sin q_2$.)

C. Derivation. *1) Potential energy and gravity torque:*

$$U = -(m_1 l_{c1} + m_2 l_1) g \cos q_1 - m_2 l_{c2} g \cos\varphi = -14.715\cos q_1 - 9.81\cos\varphi, \qquad \tau_g = -\frac{\partial U}{\partial q} = -9.81\begin{bmatrix} 1.5\sin q_1 + \sin\varphi \\ \sin\varphi \end{bmatrix}. \tag{2}$$

2) Kinetic energy and inertia matrix: the velocity of centre 2 is the elbow velocity ($l_1\dot{q}_1$) plus the rotation about the elbow ($l_{c2}\dot{\varphi}$), at angle q_2 apart (law of cosines):

$$T = \tfrac{1}{2}(m_1 l_{c1}^2 + I_1)\dot{q}_1^2 + \tfrac{1}{2} I_2 \dot{\varphi}^2 + \tfrac{1}{2} m_2 \left(l_1^2\dot{q}_1^2 + l_{c2}^2\dot{\varphi}^2 + 2 l_1 l_{c2} c_2 \dot{q}_1\dot{\varphi}\right) = \tfrac{1}{2}\dot{q}^T M \dot{q}, \quad M(q) = \begin{bmatrix} 2.663 + 2c_2 & 1.33 + c_2 \\ 1.33 + c_2 & 1.33 \end{bmatrix}. \tag{3}$$

3) Coriolis term (Euler–Lagrange): with $\partial T/\partial\dot{q} = M\dot{q}$ and $\dot{M} = \frac{\partial M}{\partial q_2}\dot{q}_2 = -s_2\dot{q}_2\left[\begin{smallmatrix} 2 & 1 \\ 1 & 0 \end{smallmatrix}\right]$,

$$\frac{d}{dt}\frac{\partial T}{\partial\dot{q}} - \frac{\partial T}{\partial q} = \tau_g + B\tau \ \Rightarrow\ M\ddot{q} + \underbrace{\left(\dot{M}\dot{q} - \frac{\partial T}{\partial q}\right)}_{=C\dot{q}} = \tau_g + B\tau, \qquad C(q,\dot{q}) = s_2\begin{bmatrix} -2\dot{q}_2 & -\dot{q}_2 \\ \dot{q}_1 & 0 \end{bmatrix}. \tag{4}$$

These are the same τ_g, M and C as in the given model.

D. State-Space Form. M is always invertible, so solving (1) for $\ddot{q}$ gives the state equation:

$$\det M = (2.663 + 2c_2)(1.33) - (1.33 + c_2)^2 = 1.7729 - c_2^2 \ge 0.7729 > 0 \ \ (c_2^2 \le 1), \qquad \dot{x} = f(x,\tau) = \begin{bmatrix} \dot{q} \\ M^{-1}\left(\tau_g + B\tau - C\dot{q}\right) \end{bmatrix}. \tag{5}$$

II. Equilibria

A. Conditions. At rest $\dot{q} = \ddot{q} = 0$, and the input torque is given as $\tau^* = 0$. Then (1) gives $\tau_g(q^*) = 0$:

$$\text{row 2: } \sin(q_1^* + q_2^*) = 0, \qquad \text{row 1: } 1.5 \sin q_1^* + \underbrace{\sin(q_1^* + q_2^*)}_{=0} = 0 \;\Rightarrow\; \sin q_1^* = 0.$$

B. Solutions. $q_1^* \in \{0, \pi\}$, $q_1^* + q_2^* \in \{0, \pi\} \Rightarrow q_2^* \in \{0, \pi\} \pmod{2\pi}$: **four equilibria**, all with both links vertical (Table I).

Table I. Equilibria for $\tau^* = 0$

	Posture	(q_1^*, q_2^*)	U [J]	Description
E_1	down–down	$(0, 0)$	-24.525	both links hanging
E_2	down–up	$(0, \pi)$	-4.905	link 1 down, link 2 up
E_3	up–up	$(\pi, 0)$	$+24.525$	fully upright (target)
E_4	up–down	(π, π)	$+4.905$	link 1 up, link 2 down

III. Linearization

A. Deviation Variables. $\tilde{q} = [q_1 - q_1^*,\ q_2 - q_2^*]^T$, $\tilde{x} = [\tilde{q}; \dot{\tilde{q}}] = x - x^*$, $\tilde{\tau} = \tau - \tau^* = \tau$. First-order Taylor expansion of (5) about $(x^*, 0)$, using $f(x^*, 0) = 0$: $\dot{\tilde{x}} \approx \frac{\partial f}{\partial x}\big|_* \tilde{x} + \frac{\partial f}{\partial \tau}\big|_* \tilde{\tau}$.

B. Jacobians. *Top half:* $\frac{d}{dt}\tilde{q} = \dot{\tilde{q}} \Rightarrow$ rows $[0\ \ I_2]$. *Bottom half:* $h = M^{-1}(\tau_g + B\tau - C\dot{q})$; at $(x^*, 0)$, $\partial(C\dot{q})/\partial\dot{q} = 0$ ($C\dot{q}$ is quadratic in $\dot{q}$, $\dot{q}^* = 0$) and $(\partial M^{-1}/\partial q)(\tau_g + B\tau) = 0$ ($\tau_g(q^*) = 0$, $\tau^* = 0$). Hence

$$\ddot{\tilde{q}} = P\tilde{q} + r\tilde{\tau}, \quad P = M(q^*)^{-1}K(q^*), \quad r = M(q^*)^{-1}B, \qquad K(q^*) = \frac{\partial \tau_g}{\partial q}\Big|_{q^*} = -9.81 \begin{bmatrix} 1.5\cos q_1^* + \cos\varphi^* & \cos\varphi^* \\ \cos\varphi^* & \cos\varphi^* \end{bmatrix}. \tag{6}$$

C. Linear State-Space Model. Stacking both halves, with the joint angles as outputs, $y = \tilde{q} = [q_1 - q_1^*,\ q_2 - q_2^*]^T$:

$$\boxed{\dot{\tilde{x}} = A\tilde{x} + B_\ell \tilde{\tau}, \quad y = C_y\tilde{x} + D\tilde{\tau}, \qquad A = \begin{bmatrix} 0 & I_2 \\ P & 0 \end{bmatrix},\ B_\ell = \begin{bmatrix} 0 \\ r \end{bmatrix},\ C_y = \begin{bmatrix} I_2 & 0 \end{bmatrix},\ D = 0} \tag{7}$$

D. Example: $E_1 = (0, 0)$. $\cos q_1^* = \cos\varphi^* = c_2 = 1$:

$$K = \begin{bmatrix} -24.525 & -9.81 \\ -9.81 & -9.81 \end{bmatrix}, \quad M = \begin{bmatrix} 4.663 & 2.33 \\ 2.33 & 1.33 \end{bmatrix}, \quad \det M = 0.7729, \quad M^{-1} = \frac{1}{0.7729}\begin{bmatrix} 1.33 & -2.33 \\ -2.33 & 4.663 \end{bmatrix},$$

$$P_{E1} = M^{-1}K = \begin{bmatrix} -12.6292 & 12.6926 \\ 14.7488 & -29.6119 \end{bmatrix}, \qquad r_{E1} = \begin{bmatrix} -3.0147 \\ 6.0332 \end{bmatrix}.$$

E. All Equilibria. M depends only on q_2^*; K only on the signs of $\cos q_1^*$, $\cos\varphi^*$. Upside down, gravity flips: $K_{E3} = -K_{E1}$ with $M_{E3} = M_{E1}$, so $P_{E3} = -P_{E1}$; likewise $P_{E4} = -P_{E2}$. Results in Table II; with (7) they give (A, B_ℓ, C_y, D) at each equilibrium.

Table II. Linearization results ($\det M = 0.7729$ at all four)

	$(\cos q_1^*, \cos\varphi^*)$	$M(q^*)$	$P = M^{-1}K$	$r = M^{-1}B$
E_1	$(1, 1)$	$\begin{bmatrix} 4.663 & 2.33 \\ 2.33 & 1.33 \end{bmatrix}$	$\begin{bmatrix} -12.6292 & 12.6926 \\ 14.7488 & -29.6119 \end{bmatrix}$	$\begin{bmatrix} -3.0147 \\ 6.0332 \end{bmatrix}$
E_2	$(1, -1)$	$\begin{bmatrix} 0.663 & 0.33 \\ 0.33 & 1.33 \end{bmatrix}$	$\begin{bmatrix} -12.6292 & 12.6926 \\ 10.5095 & 4.2266 \end{bmatrix}$	$\begin{bmatrix} -0.4270 \\ 0.8578 \end{bmatrix}$
E_3	$(-1, -1)$	as E_1	$-P_{E1}$	r_{E1}
E_4	$(-1, 1)$	as E_2	$-P_{E2}$	r_{E2}

IV. Transfer Functions

A. Laplace Transform. With $\tilde{Q}_i(s) = \mathscr{L}\{\tilde{q}_i(t)\}$, $\tilde{\mathscr{T}}(s) = \mathscr{L}\{\tilde{\tau}(t)\}$, $\tilde{Q} = [\tilde{Q}_1, \tilde{Q}_2]^T$ and zero initial conditions ($\ddot{\tilde{q}} \to s^2\tilde{Q}$):

$$s^2\tilde{Q} = P\tilde{Q} + r\tilde{\mathscr{T}} \;\Rightarrow\; (s^2 I - P)\,\tilde{Q}(s) = r\,\tilde{\mathscr{T}}(s), \qquad (s^2 I - P)^{-1} = \frac{1}{\Delta(s)}\begin{bmatrix} s^2 - P_{22} & P_{12} \\ P_{21} & s^2 - P_{11} \end{bmatrix}, \quad \Delta(s) = s^4 - \operatorname{tr}(P)\,s^2 + \det P.$$

B. General Formula. Multiplying by r and dividing each row by $\tilde{\mathscr{T}}$ ($G_i = \tilde{Q}_i/\tilde{\mathscr{T}}$):

$$\boxed{G_1(s) = \frac{r_1 s^2 + (P_{12}r_2 - P_{22}r_1)}{\Delta(s)}, \qquad G_2(s) = \frac{r_2 s^2 + (P_{21}r_1 - P_{11}r_2)}{\Delta(s)}} \tag{8}$$

Poles: roots of Δ (= eigenvalues of A); with $\lambda = s^2$, $\lambda^2 - \mathrm{tr}(P)\lambda + \det P = 0$, then $s = \pm\sqrt{\lambda}$. *Zeros:* $s^2 = -(P_{12}r_2 - P_{22}r_1)/r_1$ for G_1 and $s^2 = -(P_{21}r_1 - P_{11}r_2)/r_2$ for G_2.

C. Example: E_1. $\mathrm{tr}\,P = -42.2410$, $\det P = 186.7719 \Rightarrow \lambda = s^2 = -5.018, -37.224 \Rightarrow$ poles $\pm j2.2400$, $\pm j6.1011$; numerator constants $P_{12}r_2 - P_{22}r_1 = -12.6926$, $P_{21}r_1 - P_{11}r_2 = 31.7316 \Rightarrow G_1 = -3.0147\,(s^2 + 4.2103)/\Delta$, $G_2 = 6.0332\,(s^2 + 5.2595)/\Delta$ (row E_1 of Table III).

Table III. Transfer functions $G_i(s) = N_i(s)/\Delta(s)$ at the four equilibria

	Denominator $\Delta(s)$ (monic)	$N_1(s)$ $(\tau \to q_1)$	$N_2(s)$ $(\tau \to q_2)$	Poles	Zeros of G_1	Zeros of G_2
E_1	$s^4 + 42.2410\,s^2 + 186.7719$	$-3.0147\,(s^2 + 4.2103)$	$6.0332\,(s^2 + 5.2595)$	$\pm j2.2400$, $\pm j6.1011$	$\pm j2.0519$	$\pm j2.2934$
E_2	$s^4 + 8.4025\,s^2 - 186.7719$	$-0.4270\,(s^2 - 29.7273)$	$0.8578\,(s^2 + 7.3982)$	± 3.1775, $\pm j4.3010$	± 5.4523	$\pm j2.7200$
E_3	$s^4 - 42.2410\,s^2 + 186.7719$	$-3.0147\,(s^2 - 4.2103)$	$6.0332\,(s^2 - 5.2595)$	± 2.2400, ± 6.1011	± 2.0519	$\pm j2.2934$
E_4	$s^4 - 8.4025\,s^2 - 186.7719$	$-0.4270\,(s^2 + 29.7273)$	$0.8578\,(s^2 - 7.3982)$	± 4.3010, $\pm j3.1775$	$\pm j5.4523$	± 2.7200

D. All Equilibria. Table III. With no friction only even powers of s occur, so poles and zeros come in pairs $\pm p$; since $P_{E3} = -P_{E1}$, each $\pm j\omega$ at E_1 becomes $\pm\omega$ at E_3 (likewise E_2/E_4).

V. Verification of the Model (Phase 1)

- The model derivation and all values in Tables II and III were verified in MATLAB (agreement to four decimals).
- Nonlinear (5) vs. linear (7), from the same small deviation with $\tau = 0$ (Fig. 2): near E_1 from $\tilde{q}(0) = (0.1, -0.1)$ rad they agree within 1.0° (q_1) and 2.0° (q_2) over 8 s. Near E_3 from $\tilde{q}_1(0) = 10^{-3}$ rad the error is 0.02% when $|\tilde{q}_1| = 1^\circ$.
- With $\tau = 0$, $T + U$ stays constant to 10^{-8} (no friction).
- Phase 1 animation: **watch the video**.

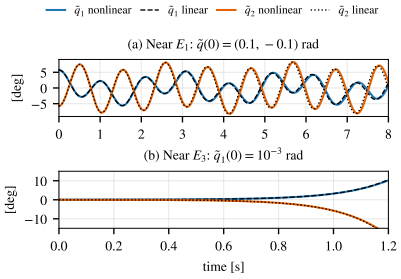


Fig. 2. Nonlinear (colour) vs. linear (black) responses near (a) E_1, (b) E_3.

VI. Stability: Criteria (Phase 2)

A. Two Views of the Same Poles. *State space:* the linear model (7) is asymptotically stable iff every eigenvalue of A has $\mathrm{Re} < 0$. It is unstable if any eigenvalue has $\mathrm{Re} > 0$. It is marginally stable (bounded, non-decaying) if the eigenvalues on the imaginary axis are simple and none lies to the right of it. *Transfer functions:* G_1, G_2 in (8) are BIBO stable iff every pole has $\mathrm{Re} < 0$. *Nonlinear system:* by Lyapunov's indirect method, an eigenvalue with $\mathrm{Re} > 0$ makes the equilibrium of (5) unstable, while eigenvalues on the imaginary axis leave the question open [5].

The two views give the same poles here. At every equilibrium $\mathrm{rank}[B_\ell\, AB_\ell\, A^2B_\ell\, A^3B_\ell] = 4$ (controllable), and (A, C_j) with $C_j = e_j^T$ has observability rank 4 for $j = 1$ and $j = 2$. Each G_j is therefore a minimal realization, with no pole–zero cancellation (in Table III no zero equals a pole). So the poles of G_1 and G_2 are exactly the eigenvalues of A, and the transfer-function verdict and the state-space verdict must coincide.

B. What the Characteristic Polynomial Allows. $\det(sI - A) = \Delta(s) = s^4 - \mathrm{tr}(P)\,s^2 + \det P$ has no s^3 and no s^1 term, because the model has no friction. All coefficients of a Hurwitz polynomial are positive, so **no equilibrium can be asymptotically stable with** $\tau = 0$; at best it is marginally stable. With $\lambda = s^2$ (the eigenvalues of P),

$$\lambda^2 - \mathrm{tr}(P)\,\lambda + \det P = 0: \qquad \lambda < 0 \Rightarrow s = \pm j\sqrt{-\lambda}\ \text{(oscillation)}, \qquad \lambda > 0 \Rightarrow s = \pm\sqrt{\lambda}\ \text{(one pole in the RHP)}. \tag{9}$$

Hence $\det P < 0$ gives one λ of each sign (one RHP pole). $\det P > 0$ with $\mathrm{tr}\,P < 0$ gives two negative λ (all poles on the $j\omega$-axis). $\det P > 0$ with $\mathrm{tr}\,P > 0$ gives two positive λ (two RHP poles).

C. Link to the Potential Energy. $K = \partial\tau_g/\partial q = -\nabla^2 U$ (the Hessian of (2)) and $M \succ 0$. $P = M^{-1}K$ is similar to the symmetric matrix $M^{-1/2}KM^{-1/2}$, which is congruent to K. By Sylvester's law of inertia, P has as many positive eigenvalues as $\nabla^2 U$ has negative ones [3]. So **the number of RHP poles equals the number of directions in which** U **decreases**: a minimum of U gives none, a saddle one, a maximum two.

VII. Stability of Each Equilibrium

Table IV. Stability of the four equilibria from the transfer functions (G_1, G_2), the state matrix (A) and the energy

	$\operatorname{tr} P$, $\det P$	$\lambda = \operatorname{eig} P$	poles of $G_1, G_2 = \operatorname{eig}(A)$	$\operatorname{eig} \nabla^2 U$	U at q^*	RHP	verdict (linear / nonlinear)
E_1	$-42.241,\ 186.772$	$-5.018,\ -37.224$	$\pm j2.2400,\ \pm j6.1011$	$4.905,\ 29.43$	minimum	0	marginally stable / Lyapunov stable
E_2	$-8.403,\ -186.772$	$-18.499,\ 10.096$	$\pm 3.1775,\ \pm j4.3010$	$-14.715,\ 9.81$	saddle	1	unstable / unstable
E_3	$42.241,\ 186.772$	$5.018,\ 37.224$	$\pm 2.2400,\ \pm 6.1011$	$-29.43,\ -4.905$	maximum	2	unstable / unstable
E_4	$8.403,\ -186.772$	$-10.096,\ 18.499$	$\pm 4.3010,\ \pm j3.1775$	$-9.81,\ 14.715$	saddle	1	unstable / unstable

A. E_1 ***(down–down): Marginally Stable, Lyapunov Stable.*** *Transfer functions:* the poles $\pm j2.2400$ and $\pm j6.1011$ are simple and on the imaginary axis, so G_1 and G_2 are *not* BIBO stable. A bounded input at a resonance, e.g. $\tau = \sin(2.24t)$, produces a response that grows like $t\sin(2.24t)$. *State space:* the same eigenvalues make the free response a bounded, undamped sum of two modes, so E_1 is marginally stable, not asymptotically stable. *Nonlinear:* linearization is inconclusive here, but energy settles it. Take $V(x) = \frac{1}{2}\dot{q}^T M\dot{q} + U(q) - U(q^*)$. It is positive definite near E_1 because $\nabla^2 U \succ 0$ there (eigenvalues $4.905, 29.43$). Along (1), $\dot{V} = \dot{q}^T B\tau = 0$ for $\tau = 0$. So E_1 is **stable in the sense of Lyapunov**, and not asymptotically stable: V is constant, so the swing never decays.

B. E_2 ***and*** E_4***: Saddles, One RHP Pole Each.*** $\det P = -186.772 < 0$, so by (9) each has one real RHP pole ($+3.1775$ at E_2, $+4.3010$ at E_4) plus an undamped pair ($\pm j4.3010$, $\pm j3.1775$). Both G_j are BIBO unstable and A is unstable. By Lyapunov's indirect method both nonlinear equilibria are unstable. U is a saddle at each: one link sits up, so the energy decreases along exactly one direction.

C. E_3 ***(up–up): Maximum, Two RHP Poles.*** Gravity flips, so $P_{E3} = -P_{E1}$ and λ changes sign. The poles $\pm j2.24, \pm j6.10$ of E_1 become ± 2.2400 and ± 6.1011. With two RHP poles, E_3 is unstable (transfer function, state space and nonlinear). U has a strict maximum, so every direction leads downhill.

VIII. Simulation Evidence

A. Simulator. All Phase 2 simulations use the full nonlinear model in **PyBullet** [6] (Featherstone multibody dynamics, uniform rods, no joint damping, step 0.25 ms). It is checked against an RK4 integration of the given equations (1): with $q(0) = (0.8, -0.5)$ rad and $\tau = 3\sin 2t$, the joint angles agree within 0.52° over 5 s. The same simulator runs the linear model (7) alongside, from the same initial condition.

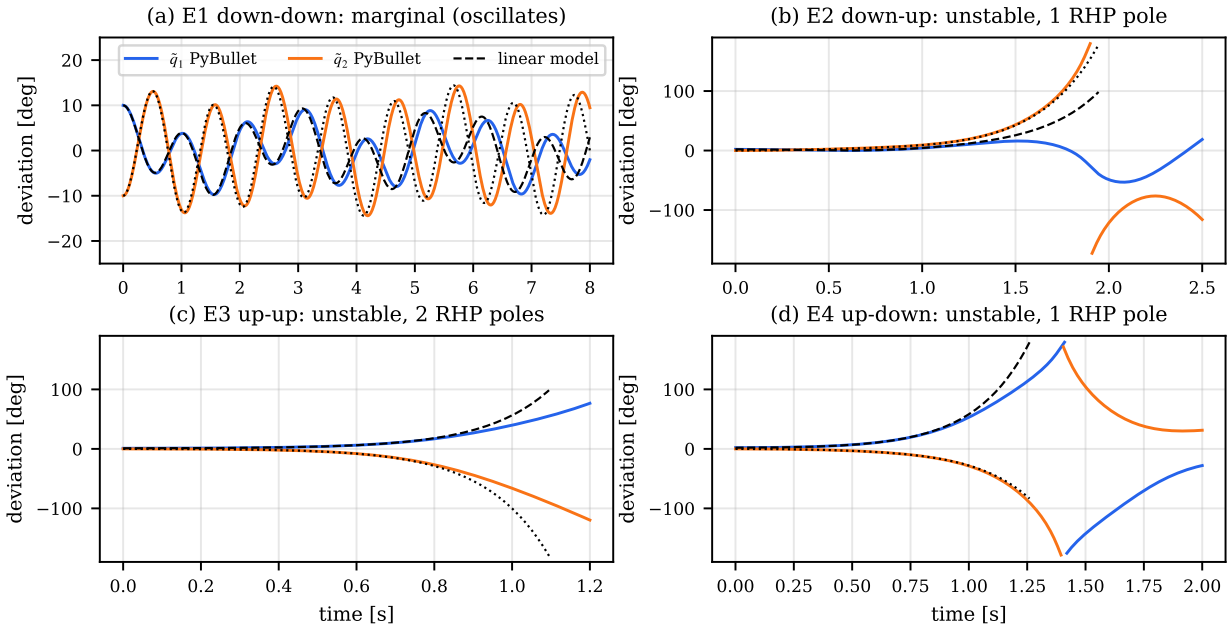


Fig. 3. Free response ($\tau = 0$) near each equilibrium: PyBullet (colour) vs. linear model (black). Initial offsets: (a) $\tilde{q}(0) = (10^\circ, -10^\circ)$; (b) 2°; (c) 1°; (d) 2° on q_1.

B. Each Conclusion Confirmed.

- E_1**:** the response stays bounded at its initial 10° amplitude and never decays (Fig. 3a), as marginal stability predicts. Over 60 s the energy changes by only 1.2×10^{-3} J, which is integration error. The spectrum of $\tilde{q}_1$ peaks at 2.239 and 5.997 rad/s (Fig. 4b). These are the poles $\pm j2.240$ and $\pm j6.101$, the second within one frequency bin ($2\pi/60 = 0.105$ rad/s) of a 60 s record.

- E_2, E_3, E_4**:** a tiny offset grows until the arm falls (Fig. 3b–d). The linear model tracks it until about 20°, then over-predicts. On a log scale the growth is a straight line (Fig. 4a). Its slope is 3.171, 6.088 and 4.294 s^{-1}, against the largest RHP poles 3.1775, 6.1011 and 4.3010 (errors 0.22%, 0.21%, 0.15%). The time to reach 30° is 2.52 s from 0.05° (E_2), 1.55 s from 0.01° (E_3) and 1.69 s from 0.05° (E_4).

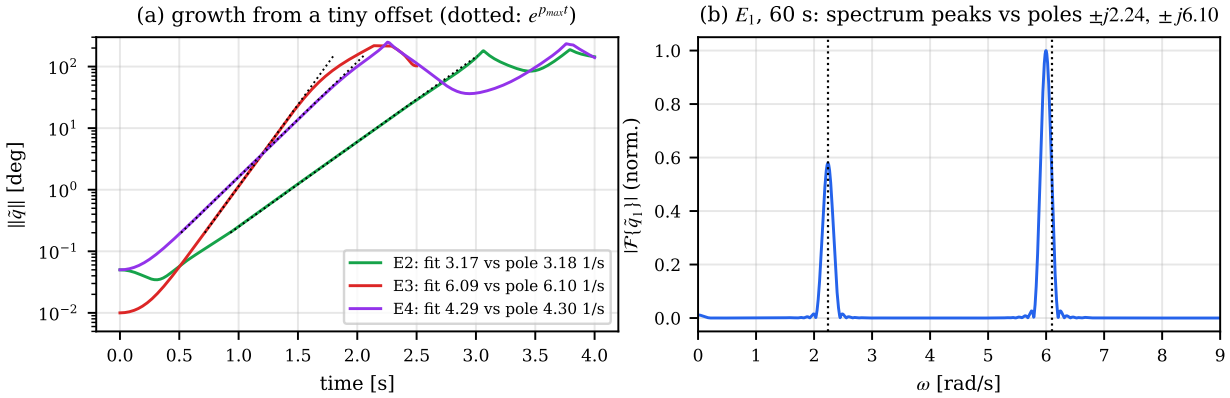


Fig. 4. (a) $\|\tilde{q}\|$ from a tiny offset; dotted lines have slope equal to the largest RHP pole. (b) Spectrum of $\tilde{q}_1$ near E_1 over 60 s; dotted lines are the poles.

IX. Single-Angle Feedback at the Upright Equilibrium E_3

A. Problem. Only one joint angle $y = \tilde{q}_j$ ($j = 1$ or 2) is measured, and the elbow torque is $\tau = -C(s)\,y$. With $G_j = N_j/\Delta$ and $C = N_c/D_c$, the closed-loop poles are the roots of $\Delta D_c + N_j N_c$. At E_3 (Table III):

$$\Delta = s^4 \underbrace{-42.241}_{a_2} s^2 + \underbrace{186.772}_{a_0}, \qquad N_1 = \underbrace{-3.0147}_{b_2} s^2 + \underbrace{12.6926}_{b_0}\ (\text{zeros } \pm 2.0519), \qquad N_2 = \underbrace{6.0332}_{b_2} s^2 \underbrace{-31.7316}_{b_0}\ (\text{zeros } \pm 2.2934). \tag{10}$$

B. Static or PD Feedback Cannot Stabilize It. With $C(s) = K_p + K_d s$ the closed-loop polynomial is

$$s^4 + (K_d b_2)\, s^3 + (a_2 + K_p b_2)\, s^2 + (K_d b_0)\, s + (a_0 + K_p b_0). \tag{11}$$

Hurwitz requires every coefficient to be positive, so $K_d b_2 > 0$ *and* $K_d b_0 > 0$, i.e. b_2 and b_0 must have the same sign. For both angles they have opposite signs ($b_0/b_2 = -z^2$), **because the zeros $\pm z$ are real: one lies in the RHP**. No K_p, K_d can satisfy both conditions. Pure proportional feedback ($K_d = 0$) is worse still: (11) is even in s, so its roots are symmetric about the imaginary axis and can never all lie in the LHP.

Root-locus view: at low gain the closed-loop poles sit near the open-loop RHP poles 2.24 and 6.10. As the gain grows they move to the zeros of $(K_p + K_d s)N_j$, one of which is the RHP zero $+z$. So every gain leaves a pole in the RHP.

A grid search over $K_p \in [-400, 400]$, $K_d \in [-150, 150]$ (Fig. 5a,b) confirms this. The smallest attainable largest real part is 0.96 s^{-1} for q_1 (at $K_p = -15$, $K_d = 0$) and 2.29 for q_2. The latter is approached at high gain, where a pole tends to the RHP zero 2.2934. In PyBullet even these best gains let a 1° offset reach 30° in 1.0 s (q_1) and 1.8 s (q_2) (Fig. 5c).

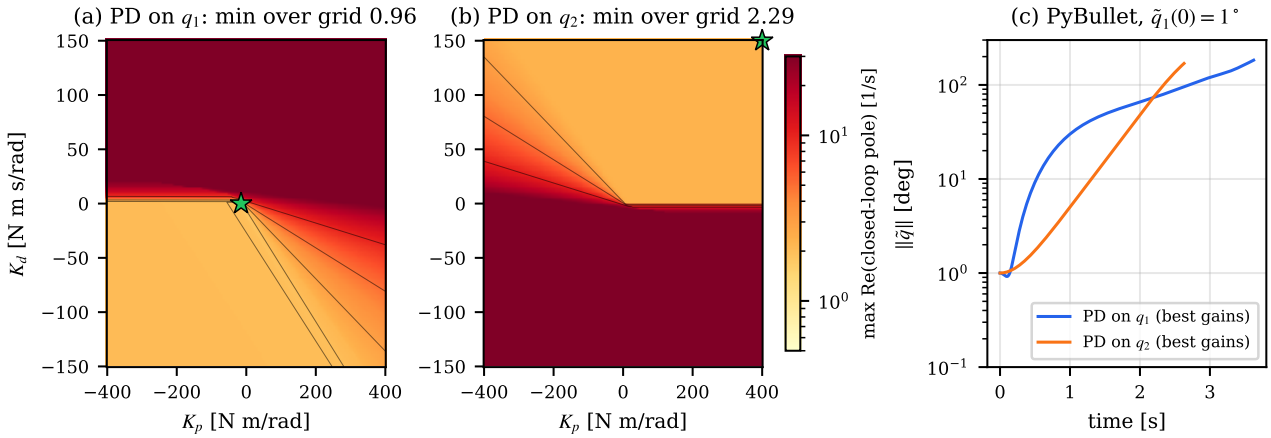


Fig. 5. PD feedback from one angle at E_3. (a, b) Largest real part of the closed-loop poles over the gain plane; it is positive everywhere (star: best gains). (c) PyBullet with the best gains: the arm still falls.

C. A Dynamic (Observer-Based) Controller Can, in Principle. Since (A, C_j) is observable for both angles (Sec. VI-A), a Luenberger observer $\dot{\hat{x}} = A\hat{x} + B_\ell \tau + L(y - C_j\hat{x})$ reconstructs the whole state from one angle. Feedback $\tau = -K\hat{x}$ on the estimate then gives closed-loop poles $\mathrm{eig}(A - B_\ell K) \cup \mathrm{eig}(A - LC_j)$ (separation principle).

- *Gains:* K is from LQR with $Q = \mathrm{diag}(10, 10, 1, 1)$, $R = 1$: $K = [-266.1, -107.4, -115.0, -54.3]$, with $\mathrm{eig}(A - B_\ell K) = -10.09, -4.25, -2.24 \pm 0.04j$. L is from the dual LQR problem: observer poles $-6.34, -5.88, -2.67, -1.90$ for q_1 and $-6.56, -5.67, -2.49, -2.03$ for q_2. All are in the LHP, so the linearized loop is asymptotically stable.
- *Stable controller or not (parity interlacing [4]):* a stable stabilizing controller exists iff every interval between consecutive real RHP zeros (including ∞, relative degree 2) contains an even number of real RHP poles. For q_1, $(2.05, \infty)$ contains 2.24 and 6.10: even, so a stable controller suffices. For q_2, $(2.29, \infty)$ contains only 6.10: odd, so **any controller that stabilizes E_3 from q_2 must itself be unstable**.
- *Fundamental limit [3]:* an RHP zero z forces $S(z) = 1$, and with RHP poles p_i every stabilizing controller has $\|S\|_\infty \geq \prod_i |z + p_i|/|z - p_i|$. That is ≥ 45.9 for q_1 and ≥ 187 for q_2, because each zero sits just below the pole 2.24. Such a peak means very poor robustness: small errors or saturation destroy stability.

D. Simulation of the Observer-Based Controller. In PyBullet (Fig. 6) the controller holds E_3 from a 1° offset measuring only q_1, and from 0.1° measuring only q_2. Measuring q_1 from 3°, the torque saturates at 20 N m and the arm falls. A bisection on the initial offset gives the largest offset that is still caught:

- measuring q_1: 1.42° (offset in q_1) and 1.66° (offset in q_2)
- measuring q_2: 0.23° and 0.97°
- full-state LQR with the same K (for comparison): 4.64° and 10.06°

The ordering $q_2 < q_1 <$ full state is the one the parity-interlacing result and the sensitivity bounds predict.

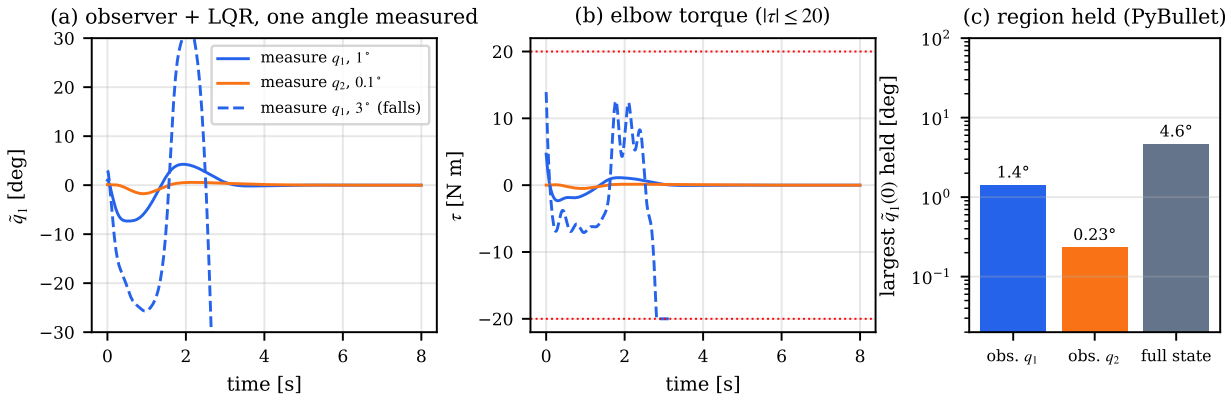


Fig. 6. Observer + LQR at E_3 in PyBullet using one measured angle. (a) $\tilde{q}_1$; (b) elbow torque; (c) largest initial offset $\tilde{q}_1(0)$ that is still stabilized.

E. Answer. **No static or PD feedback from a single joint angle can stabilize the upright equilibrium**, for either q_1 or q_2. Each G_j at E_3 has a real zero in the RHP, and that forces the sign conflict in (11): high gain drags a pole onto the RHP zero, and low

gain leaves the RHP poles in place. Stabilization from one angle **is** possible with a *dynamic* controller, because each angle alone makes the system observable. But the RHP zero sits next to the RHP pole 2.24, which makes any such loop extremely fragile. With $|\tau| \leq 20\,\mathrm{N\,m}$ it holds only $\approx 1.4^\circ$ (via q_1) or $\approx 0.2^\circ$ (via q_2, where the controller must even be unstable). Full-state feedback is the practical route for Phase 3.

X. Simulation Tools

- **Interactive PyBullet simulator**: `phase2/acrobot_phase2_sim.py`. Instructions are in `phase2/README.md`. A live dashboard lets you pick an equilibrium, an initial condition (slider, typed value, or by dragging the arm), and a controller: open loop, single-angle PD, or single-angle observer + LQR. It shows the nonlinear arm next to the linear model, $\tilde{q}(t)$, $\tau(t)$ and the closed-loop poles in the s-plane, all updating live.
- **Reproducibility**: every figure and number above is produced by `phase2deliverable/code/make_phase2_results.py`, run on the same simulator.
- **Phase 2 animation** (Manim, 2K): **watch the video**.

References. [1] M. W. Spong, "The swing up control problem for the Acrobot," *IEEE Control Syst. Mag.*, vol. 15, no. 1, pp. 49–55, 1995. [2] R. Tedrake, *Underactuated Robotics*, MIT, 2024. [Online]. Available: https://underactuated.csail.mit.edu [3] S. Skogestad and I. Postlethwaite, *Multivariable Feedback Control*, 2nd ed., Wiley, 2005. [4] D. C. Youla, J. J. Bongiorno and C. N. Lu, "Single-loop feedback-stabilization of linear multivariable dynamical plants," *Automatica*, vol. 10, pp. 159–173, 1974. [5] H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002. [6] E. Coumans and Y. Bai, "PyBullet, a Python module for physics simulation," 2016–2024. [Online]. Available: https://pybullet.org